Tasks for 03.09.2026 – Advanced Python Programming Lab

1. Using csv
   a. Write a csv file with headers: Student Name, CGPA, Certifications_Count, Internship_Exp_yrs
   b. Add 4 rows of data
   c. Print the location of the file. [Hint: absolute_path = os.path.abspath("students.csv")]
   d. Write the rows one by one
   e. Write the same data as Key Value Pairs using DictWriter and save in students_dict.csv
   f. Write another csv file demonstrating the use of header
   g. Change the mode from w to a and append three more records to the first file
   h. Read the rows from the first csv file and print them all

2. Using Pandas
   a. Load student details to a Dataframe
   b. Use to_csv() to write the the contents from a data frame to csv file
   c. Read from this csv print and print all contents using read_csv()
   d. Use the concatenation method to append few more rows to the csv file
   e. Read the updated file

3. Using json
   a. Create a dictionary of student details as in Question 1
   b. Save the dictionary as a JSON file [Hint: use json.dump()]
   c. Iterate through the JSON file and print the contents
   d. Read a dictionary as a string and convert that into a Python Dictionary object using json.load()
   e. Print the contents of the dictionary

4. Using Pandas
   a. Load a student details dictionary to a dataframe
   b. Use to_json() to save as a json file [hint: use orient]
   c. Read from the json file and print all the contents
   d. Inspect the data using head(), info(), describe() methods
   e. Access the second column and print its contents
   f. Apply filter on a column and print its contents
   g. Add a new column and load the values
   h. Demonstrate updating a value
   i. Write the updated JSON (dataframe) file